FailWhere: Can Vision-Language Models Self-Localize Their Own Failures?

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Abstract

When a Vision-Language Model (VLM) answers a visual question incorrectly, can it accurately identify which spatial region of the image caused that error? We introduce FailWhere, a benchmark that tests this capacity directly: for each answer (correct or wrong), we ask the model to name the 3×3 grid region most responsible using a structured region: tag. On controlled synthetic images (500 samples, Claude Haiku), structured self-localization accuracy is only 9% against an answer-cell ground truth, despite 75% keyword-level mention — a verbal-commitment gap showing the model hedges across regions in text but cannot commit. On real COCO photographs (VSR dataset, 142 failures), four converging lines of evidence show the region: tag is generic image re-reading rather than failure self-knowledge: **(1)** hit rates on failures match correct answers within 1.1 pp (TOST equivalence confirmed) — correctness-invariant spatial awareness; **(2)** a different model (Sonnet-5-5) that never produced the wrong answers attributes Haiku's failures equally well (+25.4 pp above null, p=0.002) — no privileged access to the answering model's state; **(3)** center tags carry zero causal signal; and **(4)** non-center tags land +23–25 pp above causal ground truth across models. Verbal self-attribution is prior-dominated in the modal case (∼54% center) but meaningfully informative when the model commits to a specific non-center region. We release the benchmark and all code.

Keywords: vision-language models, interpretability, failure attribution, benchmarks, spatial reasoning, faithfulness

1 Introduction

Modern VLMs achieve strong performance on visual question answering benchmarks, yet failures remain common on spatial reasoning tasks. Fixing these failures requires understanding *why* a model failed — which region of the image triggered the wrong answer. This is an open problem: prior attribution work either uses model internals (attention maps, gradients) or focuses on cases where the model was correct.

The gap. We map prior work onto a 2×2 table (Table 1) across two dimensions: (1) correct or wrong answer, and (2) verbal (black-box) or internal (white-box) attribution. FailWhere occupies the empty cell: verbal, black-box, failure-case localization.

Contributions. **(1)** We formalize *VLM failure self-localization*: given a failure, can the model name the responsible image region using only its verbal output? **(2)** We introduce FailWhere: a synthetic benchmark with controlled 3×3 grid images and counterfactual-masking ground truth (CF-GT). **(3)** We provide the first empirical results: 9% self-localization

	Correct answers	**Wrong answers**
Verbal	CRAFT (Parcalabescu and Frank, 2023)	FailWhere (ours)
Internal	EDCT-Bench (Zang et al., 2026); GLIMPSE	MLLMs-Know (Zhang et al., 2025)

Table 1: 2×2 positioning of FailWhere among prior attribution work. The verbal×failure cell was previously empty.

on synthetic images; a two-regime structure on real COCO photos; and a Sonnet-vs-Haiku control that resolves the "self" question cleanly. **(4)** We release benchmark code and all result files.[1]

2 Related Work

White-box failure attribution. Zhang et al. (2025) use attention maps and gradient overlays to identify which image regions VLMs attend to when making errors. FailWhere asks whether verbal self-reports approximate this signal without model internals.

Verbal correct attribution. CRAFT (Parcalabescu and Frank, 2023) tests whether VLMs explain their *correct* answers faithfully. FailWhere asks the same question on failures, where no correct explanation is available to copy from.

VLM self-knowledge and CoT faithfulness. Madsen et al. (2024) show that chain-of-thought explanations are often unfaithful post-hoc rationalizations. Our verbal-commitment gap (75% keyword vs. 9% structured accuracy) is a spatial-domain instance of this finding. Kadavath et al. (2022) show LLMs can self-calibrate factual confidence but struggle with spatial confidence. Shen et al. (2023) find VLM spatial descriptions less faithful than factual ones. Seeing-but-Not-Believing (Anonymous, 2025) shows VLMs can encode a scene correctly yet fail to believe their own percept — directly motivating our question. CLIP Cross-Eyed (Anonymous, 2026) finds that verbalized spatial descriptions misalign with internal spatial representations, consistent with our center-prior finding: the verbal channel reports a distributional prior, not an introspective readout.

3 The FailWhere Benchmark

3.1 Synthetic Benchmark

Images are 300×300 px, divided into a uniform 3×3 grid. Each cell contains one object (shape × color). Two question types:

- **Adjacent-distractor**: same shape in target and adjacent cell, different colors; model answers with the distractor color.
- **Relative-position**: "What is [dir] of the [ref-object]?" — requires two-step spatial reasoning; model misidentifies reference or shifts one cell.

1. Code, data, and result JSONs: `https://github.com/subodhwandile/vlm-failure-bench` (MIT license; maintained via the repository issue tracker).

Metric	Value
Samples	500
Error rate	8.8% (44/500)
Structured tag accuracy (answer-cell GT)	**9%** (4/44)
Keyword match (region mentioned in text)	75% (33/44)
Mean IoU (structured tag)	0.091
Failures by type (relative-pos / adj-distract)	40 / 4

Table 2: Synthetic benchmark results (Claude Haiku, n=500, temp= 0).

Ground truth: *answer-cell GT* (cell containing the correct answer).

3.2 Real-Image Extension: VSR

We use the Visual Spatial Reasoning dataset (Liu et al., 2023) (VSR, `cambridgeltl/vsr_random`, split `test`), which pairs COCO train2017 photographs with 66 spatial-relation caption types. Task: classify each caption as True or False. We use the first 500 items, at temperature = 0.0.

3.3 Counterfactual Masking Ground Truth (CF-GT)

For each failure, we mask each of 9 cells with solid gray (128,128,128) and re-run the VLM. The cell whose masking corrects the wrong answer is CF-GT. If no cell fixes it: `none_found`. If several cells each fix it: `multiple`. Only single-cell *clean* outcomes (n=22/142, 15.5%) are used to score tag accuracy; this 15.5% rate is the hard attribution ceiling for this dataset.

3.4 Self-Localization Protocol

Three prompt conditions (full templates in Appendix A):

- **Condition A (Neutral)**: "Which region most influenced your answer?" — all items.
- **Condition B (Stateless)**: "Your answer was wrong. Which region caused it?" — failures only.
- **Condition C (Multi-turn)**: original T/F exchange prepended as context before attribution prompt — failures only.

The REGION: tag uses exactly one of nine labels (top-left, top-center, ..., bottom-right). Parse rate: 100% on the VSR run.

4 Results

4.1 Synthetic Benchmark

Table 2 shows a *verbal-commitment gap*: 75% keyword mention vs. 9% structured accuracy. The model mentions the correct region in its explanation but commits to a different one in the REGION: tag — hedging across many regions in free text while defaulting to a wrong

	Correct (n=336)	Failed (n=131)	Δ
Subject cell hit	46.1%	45.0%	−1.1 pp
Object cell hit	26.2%	28.2%	+2.1 pp
Either cell hit	59.2%	61.8%	+2.6 pp
Baseline: uniform	11.1%	11.1%	—
Baseline: perm-null	23.2%	22.9%	—
Baseline: always-center	49.1%	49.6%	—

Table 3: Per-instance COCO bbox hit rates for neutral tag (Cond. A), comparing correct vs. failed answers (n=467 scored items, 93.4% join rate). TOST equivalence test (±8 pp bounds): p=0.86 (subject), p=0.62 (either) — cannot reject equivalence.

heuristic when forced to name one. Both the 200-sample pilot (0% / 17 failures) and this 500-sample run confirm near-zero self-localization under answer-cell GT.

4.2 Real Images: Center Bias

On 142 failures (VSR, temp= 0), the model tags "center" in 45.8% of Condition B responses — 4.1× the uniform prior (11.1%). The per-instance COCO bbox analysis (Section 4.3) shows this reflects veridical image statistics: COCO subjects are in the center cell 49.3% of the time. The center-bias is not an error artifact — it tracks object distributions.

4.3 Correctness-Invariant Spatial Awareness

Table 3 shows three decisive findings:

(1) Hit rates on failed answers match correct answers within 2.6 pp (all within 95% Wilson CIs). TOST equivalence (±8 pp) is confirmed ($p \geq 0.62$): the REGION: tag does not distinguish whether the answer was correct.

(2) The verbal report is prior-dominated: "always predict center" achieves 49.1–49.6% subject-hit, *beating* the model's 46.1–45.0%. The REGION: tag captures image-object priors but adds no discriminative information beyond them.

(3) Center sensitivity analysis: when the subject IS in center, the model tags center 57.6% (correct) / 53.8% (failed); when it is not, the model still tags center 25.1% / 31.8%. As a binary center-detector: sensitivity ≈56%, FPR ≈28%, AUC ≈0.64 — a weak detector.

4.4 Counterfactual Attribution: Two-Regime Structure

Two-regime model. Table 4 reveals a clean bifurcation:

Center tags carry zero causal signal. The center hit rate equals the permuted-null and the always-center baseline (both ≈14%); excess is 0 pp across all three conditions. Center self-attribution provides no information about what actually caused the error.

Non-center tags carry real signal. When the model escapes the center prior, it hits CF-GT at 26.3% vs. a 17.9% null — a consistent +16 pp excess across all three conditions, approximately 2.3× above chance.

	Neutral (A)	Stateless (B)	Multi-turn (C)
Clean hit rate (n=22)	27.3%	31.8%	36.4%
Perm-null	13.5%	14.0%	13.7%
Excess over null	+13.8 pp	+17.8 pp	+22.7 pp
Tag = center (n=3)	33.3%	66.7%	33.3%
excess	+0.0 pp	+0.0 pp	+0.0 pp
Tag ≠ center (n=19)	26.3%	26.3%	26.3%
excess	+14.9 pp	+16.4 pp	+16.4 pp

Table 4: Tag vs. CF-GT accuracy on clean single-cell subset (n=22). Center excess is exactly null across all conditions; non-center excess is stable at +15–16 pp — identical across conditions.

	Sonnet-5-5		Haiku-4-5	
	Neutral	Stateless	Neutral	Stateless
Clean hit (n=22)	31.8%	**36.4%**	27.3%	31.8%
Excess over null	+20.4 pp	+25.4 pp	—	—
p-value (one-sided)	0.007	**0.002**	—	—
Center bias	39.7%	41.4%	41.8%	45.4%

Table 5: Sonnet-5-5 attributing Haiku-4-5's 142 VSR failures vs. Haiku's own attributions, scored against Haiku's CF-GT. Sonnet matches or slightly exceeds Haiku despite never having produced the wrong answers. Sonnet stateless result is the paper's strongest significant number: +25.4 pp, p=0.002.

Context (Condition C) does not improve attribution. The non-center hit rate is identical across all three conditions (26.3%), confirming the two-regime structure is prompt-invariant. The multi-turn gain on the overall number (+22.7 pp) is a center-stratum artifact (n=3, noise).

Failure typology. Of 142 CF runs: `none_found` 51.4%, `multiple` 33.1%, clean 15.5%. Most VSR failures are causally diffuse at 3×3 granularity — no single cell explains the error. The clean ceiling is the paper's hardest constraint: any tag-accuracy claim on this dataset is bounded by 15.5%, not 100%.

Label asymmetry. True-caption failures (model said False, n=104): 60.6% none_found, 11.5% clean — holistic spatial-comprehension failures. False-caption failures (model said True, n=38): 26.3% none_found, 26.3% clean — more localized, region-specific evidence.

4.5 Other-Model Baseline: Sonnet Reads Haiku's Failures

Design. Self-knowledge control: Sonnet-5-5 is given the image, caption, and Haiku's wrong answer, then produces a REGION: tag. If Sonnet matches Haiku's attribution accuracy, the tag reflects generic image re-reading; if it is worse, self-knowledge is required.

Result. Table 5 shows Sonnet matches or slightly exceeds Haiku: 31.8–36.4% vs. 27.3–31.8%. The Sonnet stateless result is the paper's strongest significant number: +25.4 pp above null, p=0.002. A model that never produced the wrong answer attributes the failure as well as the model that did.

Implication. The region: tag is a generic re-reading of the image, not privileged access to the answering model's internal state. Combined with Section 4.3's correctness-invariance, Sections 4.4–4.5 jointly establish the four-pillar story of Section 5.

5 Discussion

Four pillars. The paper's results converge on a complete structured negative result:

1. **Reads the image, not the mind** — Sonnet attributes Haiku's failures equally well (+25.4 pp, p=0.002). No privileged model-state access is needed or demonstrated.

2. **Doesn't know if the answer was wrong** — identical hit rates on correct and failed answers ($\Delta = -1.1$ pp, TOST ± 8 pp confirmed).

3. **Commits reasonably when it escapes center** — non-center tags land +15–25 pp above null across conditions and models.

4. **Most reports never escape** — ∼54% are center-tagged; center tags carry zero causal signal.

The center bias is not spurious. COCO subjects sit in the center cell 49.3% of the time; the model's 42% neutral-center rate reflects real image statistics, not a failure artifact. However, the center prior is counterproductive under failure conditions: causal cells are 86% non-central (19/22 clean), so always-center achieves only 13.6% CF-GT accuracy vs. the model's 27–32%. The center prior is informative for object localization and harmful for failure attribution — the two tasks are confounded only at the level of the verbal tag.

Two GT definitions and their ceiling. *Answer-cell GT* (cell containing the correct answer) targets capability; *CF-GT* (cell whose masking fixes the error) targets causality. On the VSR dataset, 84.5% of failures have no clean CF-GT (≥ 2 cells or none required); the effective attribution ceiling is 15.5%. Any result reporting tag-vs-CF-GT accuracy must be interpreted relative to this ceiling, not 100%.

Limitations. (1) Caption-to-COCO bbox matching uses heuristic noun extraction (93.4% join rate). (2) All experiments use Claude Haiku and Sonnet; cross-family replication (GPT-4V, Gemini) remains future work. (3) Task scope is binary T/F (VSR); generalization to open-ended VQA is untested. (4) The stateless prompt asks for post-hoc attribution; the model has no memory of its visual processing chain. (5) CF masking at 3×3 has a resolution limit: synthetic adjacent-distractor cases return `multiple` (4/4) even when the causal cell is known by construction. (6) Fix-rate secondary analysis (n=64, +4.9 pp, p=0.20) is directionally consistent with the binary result but underpowered.

6 Conclusion

We introduce FailWhere and show that VLM verbal self-localization is generic image re-reading with a center prior — not failure self-knowledge. The result is a complete structured negative: the region: tag (1) requires no model-state access (Sonnet control), (2) does not detect failure (correctness-invariance), (3) is informative when non-center (+15–25 pp), and (4) is mostly captured by the center prior (∼54% of reports). These findings motivate both better introspection training objectives and more precise evaluation frameworks that distinguish self-report from genuine failure attribution.

Broader Impact Statement

This work studies whether vision-language models can accurately identify the image region responsible for their own errors. Positively, a reliable self-localization capacity would support error diagnosis, targeted model debugging, and more trustworthy deployment of VLMs in spatial-reasoning settings (e.g., robotics, document understanding, assistive tools). Our findings are cautionary in a way that also benefits practitioners: a fluent, confident region: tag is largely generic image re-reading driven by a center prior, so downstream systems should not consume verbal failure attributions as ground-truth causal evidence. Negatively, the same finding could encourage over-trust in models that *appear* self-aware; misattributed explanations could misdirect debugging effort or mask systematic failure modes. The benchmark uses public datasets (COCO images via the VSR dataset) and synthetic images; it contains no personal data and poses no privacy concerns.

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Appendix A. Prompt Templates

All prompts use temperature = 0.0 and model `claude-haiku-4-5-20251022` (public Anthropic API) / `global.anthropic.claude-haiku-4-5-20251001-v1:0` (Bedrock). Sonnet baseline (§4.5): `claude-sonnet-5-5` / `global.anthropic.claude-sonnet-5-5`; `temperature` omitted on Bedrock (deprecated parameter for this model).

A.1 True/False Question (all conditions)

```
Is the following statement true or false?

Statement: "{caption}"

Answer with only "True" or "False".
```

A.2 Condition A — Neutral Self-Localization

```
You just answered the following question about this image:
"{caption}"

Your answer was: "{model_answer}"
```

```
Which region of the image most influenced your answer?

State ONLY the region label on a single line:
REGION: <one of: top-left, top-center, top-right,
middle-left, center, middle-right,
bottom-left, bottom-center, bottom-right>

Then briefly explain why that region influenced your answer.
```

A.3 Condition B — Stateless Failure-Specific

```
You answered the following question INCORRECTLY.

Question: Is the following statement true or false?
Statement: "{caption}"

Your answer was: "{model_answer}" (this was wrong).

Which region of the image most influenced your incorrect answer?

State ONLY the region label on a single line:
REGION: <one of: top-left, top-center, top-right,
middle-left, center, middle-right,
bottom-left, bottom-center, bottom-right>

Then explain:
1. What visual content in that region led to the wrong answer
2. Why that region caused your error
```

A.4 Condition C — Multi-Turn

```
[Turn 1 | user]
Is the following statement true or false?
Statement: "{caption}"
Answer with only "True" or "False".

[Turn 1 | assistant]
{model_answer}

[Turn 2 | user]
That answer was incorrect. Looking back at the image,
which region most influenced your incorrect answer?

State ONLY the region label on a single line:
REGION: <one of: top-left, top-center, top-right,
```

```
middle-left, center, middle-right,
bottom-left, bottom-center, bottom-right>

Then explain:
1. What visual content in that region led to the wrong answer
2. Why that region caused your error
```

A.5 Grid Cell Definitions

The image is divided into a uniform 3×3 grid:

Cell	x_0	y_0	x_1	y_1
top-left	0	0	1/3	1/3
top-center	1/3	0	2/3	1/3
top-right	2/3	0	1	1/3
middle-left	0	1/3	1/3	2/3
center	1/3	1/3	2/3	2/3
middle-right	2/3	1/3	1	2/3
bottom-left	0	2/3	1/3	1
bottom-center	1/3	2/3	2/3	1
bottom-right	2/3	2/3	1	1

CF masking: cells are filled with solid gray (128, 128, 128). Images encoded as PNG base64 in API requests to avoid JPEG re-compression artifacts in masked regions.